

Element A

After constant confusion, disorganization, and instability issues, Dr. Paz decided it was time to modify his current materials organizer. During projects, Dr. Paz and students face disorganized and sprawling material locations, which occurs throughout every project, causing students to spend extra time looking for materials, Dr. Paz to spend time organizing and displaying materials, materials to be stored and provided in manners with noticeably higher risks to students of injuries. During the cleansing of the classroom in the post-production phase of classes on days of construction, students face struggles returning their materials in a timely manner, which occurs whenever materials are reorganized after being put away between days or projects, causing time to be wasted finding and putting away materials, delaying efforts to get to class on time. The issue affecting Dr. Paz and his students occurs in Dr. Paz's classroom, which is caused by his current model being unstable and a flimsy materials organizer. Due to this, the model has not been used. Thus, there are safety hazards when tools like blades and scissors can't be properly stored due to there not being proper containers that keep those sharp objects and their properties in mind, as well as hot glue guns that can't properly cool down between classes, leading to the possibility of burns as students reach in bins to retrieve glue guns. It is also very unstable, leading to its inefficiency. Dr. Paz hasn't used it because it can topple everything over, putting Dr. Paz in danger of getting cut from the blades or sustaining any other injury. In a study by Love et al. (2021) data shows that [SOURCE](#): "The majority of teachers (67%) reported 1-10 safety incidents occurred, with the top causes being hot glue guns (36%), student operated equipment (31%), and hand or portable power tools (18%)." (pg. 8) exemplifying how dangerous equipment such as hot glue guns can be and how there needs to be proper storage for them. Many more studies have been done to see the importance of being organized. According to Diaz (2023) from [SOURCE](#): she states that "A lack of organization may affect how much time it takes you to do things because you don't always know what you need to do, or you may not have the tools to get it done." further showing how important it is to be organized, especially in a classroom setting, where time is everyone for students and teachers. In a similar study by Beckwith and Parkhurst (2022) [SOURCE](#): they found that "If the physical space around us feels scattered, it's likely our mental space will feel the same." When working with these tools, everyone must be in a focused mindset, or else it leaves room for accidents to occur. According to Utuk et al. (2024) [SOURCE](#): it was also found that "Based on the findings of the study, it is concluded that effective application of correct hazard management practices such as proper tool storage strategies and safety instructions could lead to accident prevention in Technical College workshops in Akwa Ibom State." This importantly demonstrates how it is crucial to be organized when there are tools that can be dangerous if not stored properly, further showing how it is vital that Dr. Paz has properly stored tools.

By tackling this issue we can not only improve the layout of Dr Paz's tools and make his setup for projects more efficient but we would also improve the effectiveness of students. Having

all their resources in one place would allow them to see all the tools they have at their disposal and not have to ask Dr Paz for where certain tools are. That same aspect could also help Dr Paz keep inventory, seeing which tools/resources are running low. The general aspect of saving time would have beneficial impacts on Dr. Paz's ability to teach, giving more time before class to make lesson plans, interact with students and improve his and his student's quality of life, this would naturally reduce cortisol levels for his students. ([Health benefits of organisation](#)). The primary stakeholders for this project are Dr. Paz himself, future students who are going to use this, the school, ourselves. Primary concerns include but are not limited to: Utility(usefulness in general), Safety, Stability, Storage(space), and Usability(how user friendly is it?). Secondary stakeholders for this project, other engineering teachers, and administrations. Secondary concerns include but are not limited to: Mobility, Aesthetics(designs, and how pretty it is), Materials, and Identifiability(labels: how easy it is to know which is what, and where is such and so?).

Approximately 200 students every year and several dozen students every year, resulting in delayed leaving of class by up to 3.207 minutes, removing 64.14% of the time allotted by school policy to reach your next class. This also affects the majority of teachers as 67% of them from the southern US reported 1-10 safety incidents occurred, with the top causes being hot glue guns (36%) student operated equipment (31%), and hand or portable power tools (18%).” It is crucial to fix this problem because it is important to put student’s safety and wellbeing first, and in order to prevent any more injury from hot glue guns and other hand tools, we have to start with those tools being in appropriate and stable areas, where students won't accidentally burn their hands as they reach for a hot glue gun from a bin. As stated prior, it is important to be organized as it can create a more focused mental state, that being a benefit to students and teachers, showing how crucial it is to fix all of these problems that Dr. Paz’s old model currently has.

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